

Slides:

https://docs.google.com/presentation/d/1G-FxZxq68aSTIdIKaT6Q4DwJbR6OnKWxbCWc2BW98qw/edit?slide=id.g3f87bd6372e_1_1572#slide=id.g3f87bd6372e_1_1572

Group Case 5: PORSCHE DRIVE (B): Pricing Analytics

(HEVEN) Q1. What are the current practice cost-based pricing direct and indirect costs for a car subscription program?

Porsche Drive currently uses a cost-based pricing strategy, meaning the company determines the cost of operating the car subscription program and then adds a markup to cover those costs in order to generate a profit. When setting the subscription price, Porsche must consider both direct and indirect costs. The largest direct cost is vehicle depreciation, which can be calculated as the difference between the price Porsche pays for the vehicle and the vehicle's resale value when it is retired from the subscription fleet. Another direct cost is insurance, which is relatively expensive because Porsche vehicles are luxury vehicles. In addition to these direct costs, Porsche Drive must account for several indirect costs associated with operating the overall subscription program. These include the opportunity cost of idle vehicles; this is because a car that is sitting unused is not generating subscription revenue. Other indirect costs include administrative expenses, marketing expenses, and employee salaries required to manage and support the program. Therefore, under Porsche Drive's current cost-based pricing approach, the subscription price needs to account for the direct and indirect costs. Porsche then uses these costs as the basis for determining a subscription price that covers its expenses while allowing the company to earn a profit.

Nia Q2. What are the current practice cost-based pricing fixed and variable costs for a car subscription program?

Fixed costs are expenses that Porsche Drive generally incurs regardless of how frequently a vehicle is used or how many customers subscribe to the program. These costs are relatively stable over a given period and are necessary to maintain the fleet and operate the subscription service. These include vehicle depreciation, administrative and overhead expenses, fleet financing/capital costs, and registration and taxes. These costs are associated with maintaining the Porsche Drive fleet and operating the overall subscription program.

Variable costs are expenses that can change depending on the level of vehicle usage, the number of customers served, or the services provided during a subscription. For Porsche Drive, maintenance costs are variable because greater vehicle usage can lead to more frequent tire replacements, oil changes, brake servicing, cleaning, and other maintenance requirements. These

include maintenance costs such as tires, cleaning, oil changes, and brakes, as well as insurance, delivery/pick-up charges, roadside assistance, and damage or repair costs. Insurance costs can also be treated as variable in the context of the subscription program because they are associated with providing coverage for vehicles while they are being used by customers. Delivery and pick-up charges may vary depending on how frequently vehicles are delivered or collected from customers. Roadside assistance costs can also vary based on how often customers require the services. Damage and repair costs vary depending upon the condition of the vehicle. Porsche Drive's fixed costs are primarily associated with owning and maintaining the fleet and operating the subscription business, while variable costs are more closely related to vehicle usage and the services required by individual customers.

Julia Q3. Market-based Pricing: Should Porsche Drive reduce its prices given the current market conditions?

Porsche Drive should consider reducing its prices based on the current market conditions, but it should not make a large price reduction. The market has changed since Porsche Drive originally established its cost-based pricing strategy. New and used vehicle prices had increased during the pandemic and chip shortage, but by 2023, new vehicle inventories were rising and transaction prices were potentially falling. Interest rates had also plateaued, which could make traditional vehicle financing more attractive to some customers. These factors could put downward pressure on demand for Porsche Drive.

Competition should also be considered. Volvo was the only major luxury manufacturer that continued its subscription program during the pandemic, and its highest monthly subscription price was still lower than Porsche's lowest Macan subscription price. At the same time, Audi, BMW, Cadillac, and Mercedes-Benz had discontinued their subscription programs. This reduced the amount of direct competition Porsche Drive faced and may have contributed to its strong demand.

Therefore, Porsche Drive should **re-evaluate its pricing rather than automatically lowering prices**. The case indicates that Porsche had underestimated vehicle resale values, which may have resulted in prices being too high, even though demand was stronger than expected. A more flexible, market-responsive pricing strategy could help Porsche respond competitively to changes in the market while avoiding unnecessarily reducing revenue.

Overall, Porsche Drive should consider a **modest or targeted price reduction** while continuing to monitor competitors, demand, vehicle prices, and interest rates. This would allow Porsche to respond to changing market conditions while taking advantage of its relatively limited competition in the ultra-luxury subscription market.

Haotian Q4. Value-based Pricing: Single-car subscription value-based pricing problem.
Suppose a customer wants to access a Macan S for 12 months. Should they finance at 7.75 per cent tax rate and \$500 fee or subscribe to the vehicle, and why?

Value-Based Pricing: Macan S

Assumptions:

MSRP = \$84,185

Sales tax rate = 7.75%

Transaction fee = \$500

One-year residual value = 81% of MSRP

Annual insurance cost = \$2,279

Monthly subscription price = \$1,700

Option A: Finance the vehicle and sell it after 12 months

Sales tax:

$\$84,185 \times 7.75\% = \$6,524.34$

Total acquisition cost:

$\$84,185 + \$6,524.34 + \$500 = \$91,209.34$

Expected resale value after one year:

$\$84,185 \times 81\% = \$68,189.85$

Net vehicle cost:

$\$91,209.34 - \$68,189.85 = \$23,019.49$

Total cost after adding insurance:

$\$23,019.49 + \$2,279 = \$25,298.49$

Equivalent monthly financing cost:

$\$25,298.49 \div 12 = \$2,108.21$ per month

Option B: Subscribe for 12 months

Subscription cost before tax and fees:

$\$1,700 \times 12 = \$20,400$

Subscription cost including the 7.75% tax and \$500 fee:

$$(\$20,400 \times 1.0775) + \$500 = \$22,481$$

Comparison:

Savings from subscribing:

$$\$25,298.49 - \$22,481 = \$2,817.49$$

Conclusion:

The customer should subscribe to the Macan S. The estimated 12-month subscription cost is \$22,481, compared with \$25,298.49 for financing and then selling the vehicle. The subscription therefore saves approximately \$2,817.49. It also includes insurance, maintenance, roadside assistance, and concierge service while eliminating resale-value risk and the inconvenience of selling the vehicle.

The customer's approximate value-based price ceiling is \$2,108 per month before subscription taxes and fees. If the subscription price remains below this amount, subscribing provides greater value than financing. Because loan interest and selling costs are excluded from the financing calculation, the actual advantage of subscribing may be even greater.